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Section: AI(A)

QR Code :

```
import qrcode
```

```
from PIL import Image
```

```
data = "KHADIJA JAWAD | ABDUL WALI KHAN UNIVERSITY | Mardan | Computer Science |  
2023-2027"
```

```
qr = qrcode.QRCode(  
    version=1,  
    error_correction=qrcode.constants.ERROR_CORRECT_M,  
    box_size=10,  
    border=4  
)
```

```
qr.add_data(data)
```

```
qr.make(fit=True)
```

```
img = qr.make_image(fill_color="black", back_color="white")
```

```
img.save("qrcode.png")
```

```
print("QR code saved as qrcode.png")
```

Explanation:

The QR code generated in this assignment encodes my personal academic information including my full name, department, university ,city (Mardan), and the duration of my bachelor's degree. When scanned using any smartphone QR reader or camera app, the code instantly displays this information in plain text format. The QR code was created using Python programming with the qrcode and Pillow libraries, which allowed me to customize the size, error correction level, and appearance of the code. QR codes work by encoding data into a matrix of black and white square modules, where the pattern of these modules represents binary data that scanners can decode.